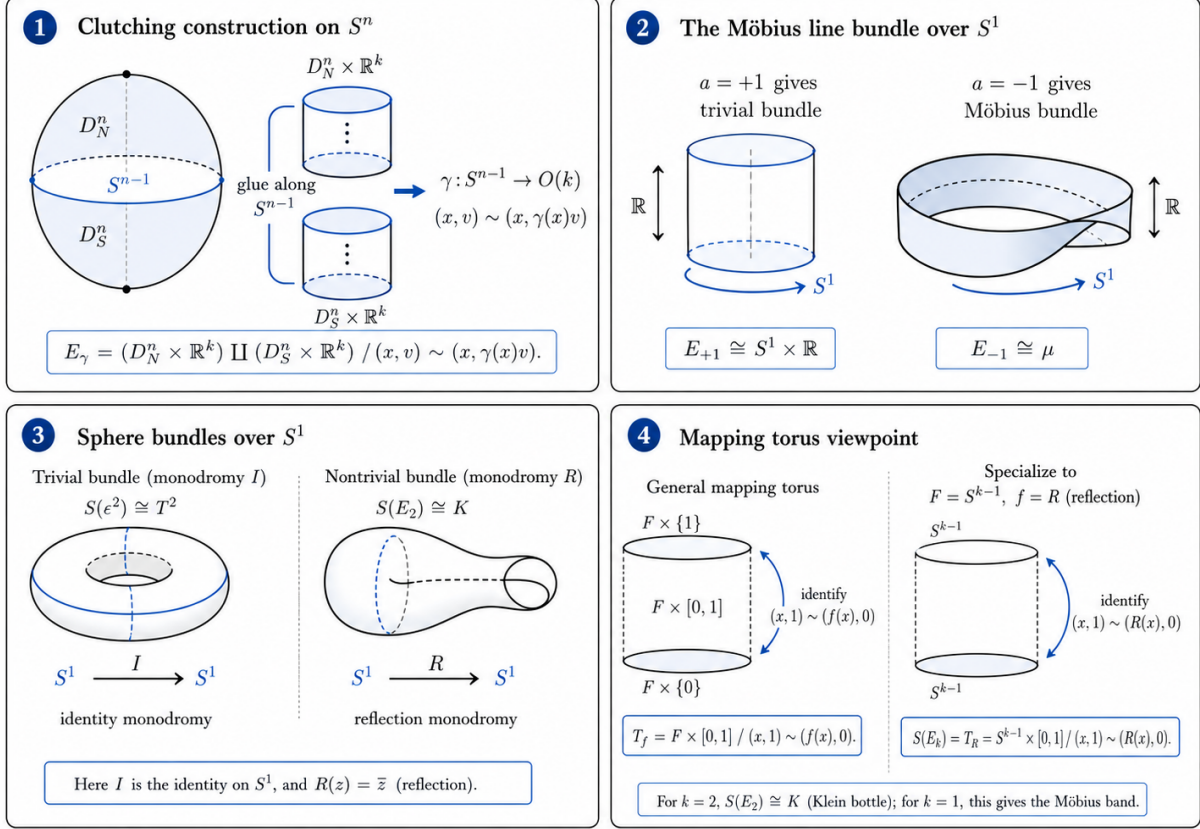


# 1 Diagrams and summary charts for Problems 11–12

The following diagrams summarize the geometric content of the clutching construction, the unique nontrivial bundle over  $S^1$ , and the associated sphere bundles. They are intended to support the solutions of Problems 11–12 by showing how the abstract gluing data is represented geometrically.

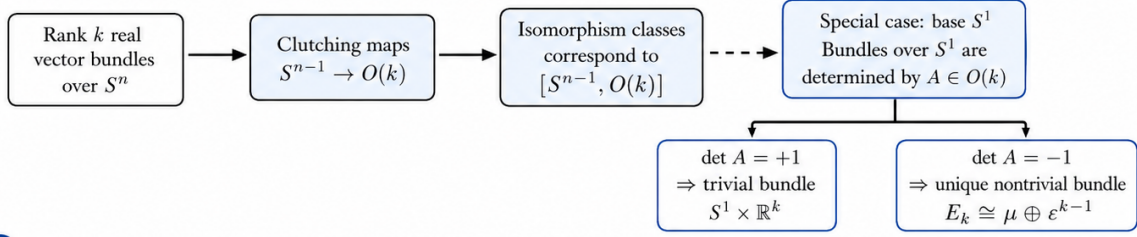
## Vector bundles over spheres: key geometric diagrams



**Figure 1:** Key geometric diagrams for vector bundles over spheres. The first panel shows the clutching construction over  $S^n$ ; the second compares the trivial line bundle over  $S^1$  with the Möbius line bundle; the third compares the torus and Klein bottle as sphere bundles over  $S^1$ ; the fourth interprets these examples as mapping tori.

## Classification and examples: vector bundles over spheres

### 1 Classification flow chart



### 2 Examples and comparison

| Base  | Rank | Clutching data                                      | Result                                                                                              |
|-------|------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| $S^1$ | 1    | $+1 \in O(1)$                                       | trivial line bundle $S^1 \times \mathbb{R}$                                                         |
| $S^1$ | 1    | $-1 \in O(1)$                                       | Möbius line bundle $\mu$                                                                            |
| $S^1$ | 2    | $I \in O(2)$ ( $\det I = +1$ )                      | sphere bundle = torus $T^2 = S^1 \times S^1$ (trivial)                                              |
| $S^1$ | 2    | $R = \text{diag}(1, -1) \in O(2)$ ( $\det R = -1$ ) | sphere bundle = Klein bottle $K$                                                                    |
| $S^2$ | 1    | $S^1 \rightarrow O(1) = \{\pm 1\}$                  | only trivial line bundle                                                                            |
| $S^2$ | 2    | degree $m$ map $S^1 \rightarrow SO(2) \cong S^1$    | disk (or sphere) bundle with Euler number $m$<br>(total space = $D(E_m)$ ; sphere bundle $S(E_m)$ ) |
| $S^2$ | 2    | degree 2 map $S^1 \rightarrow SO(2)$                | $TS^2$ (tangent bundle of $S^2$ )                                                                   |

### 3 Cohomology of $S(E_k)$ for $k \geq 2$

|                                                                                                                                                                      |                                                                                                                                                     |                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| For the unique nontrivial rank- $k$ bundle $E_k$ over $S^1$<br>( $E_k \cong \mu \oplus \varepsilon^{k-1}$ ), the cohomology of its unit<br>sphere bundle $S(E_k)$ is | $H^q(S(E_k); \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & q = 0 \\ \mathbb{Z} & q = 1 \\ \mathbb{Z}/2 & q = k \\ 0 & \text{otherwise.} \end{cases}$ | <b>Examples:</b> <ul style="list-style-type: none"> <li><math>S(E_2) \cong</math> Klein bottle <math>K</math>.</li> <li><math>S(E_3) =</math> mapping torus of a reflection of <math>S^2</math>.</li> </ul> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Figure 2:** Classification chart and examples for vector bundles over spheres. Rank  $k$  real vector bundles over  $S^n$  are described by clutching maps  $S^{n-1} \rightarrow O(k)$ . The special case of bundles over  $S^1$  reduces to the determinant of the monodromy matrix  $A \in O(k)$ . The chart also records the cohomology of the unit sphere bundle  $S(E_k)$  for the unique nontrivial bundle  $E_k \rightarrow S^1$ .

### How to read the diagrams

In Figure 1, the bundle  $E_\gamma \rightarrow S^n$  is obtained by gluing two trivial bundles over the northern and southern hemispheres. The gluing is performed along the equator  $S^{n-1}$ , and the gluing rule is encoded by the clutching map

$$\gamma : S^{n-1} \longrightarrow O(k).$$

Thus the whole bundle is determined by how the fibers are identified along the equator.

For bundles over  $S^1$ , the monodromy matrix  $A \in O(k)$  determines the bundle. If  $\det A = +1$ , the bundle is trivial. If  $\det A = -1$ , one obtains the unique nontrivial rank  $k$  bundle

$$E_k \cong \mu \oplus \varepsilon^{k-1}.$$

Its unit sphere bundle is the mapping torus of a reflection of  $S^{k-1}$ :

$$S(E_k) \cong S^{k-1} \times [0, 1] / (u, 1) \sim (Ru, 0).$$

For  $k = 2$ , this gives the Klein bottle:

$$S(E_2) \cong K.$$